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Functional homogenization of terrestrial mammals outside protected areas in the Hengduan Mountains, China

Xueyou Li  | Wenqiang Hu  | Kenneth Otieno Onditi  | Quan Li | Zhechang Hu | Ru Bai | Yang Liu | Hongjiao Wang | Xuelong Jiang

State Key Laboratory of Genetic Evolution and Animal Models & Yunnan Key Laboratory of Biodiversity and Ecological Conservation of Gaoligong Mountain, Kunming Institute of Zoology, Chinese Academy of Sciences, Kunming, China

Correspondence

Xueyou Li and Xuelong Jiang, State Key Laboratory of Genetic Evolution and Animal Models & Yunnan Key Laboratory of Biodiversity and Ecological Conservation of Gaoligong Mountain, Kunming Institute of Zoology, Chinese Academy of Sciences, No. 17 Longxin Road, Kunming, Yunnan 650201, China. Email: lixueyou@mail.kiz.ac.cn and jiangxl@mail.kiz.ac.cn

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Abstract

Despite the crucial role of human impacts on biodiversity loss, many assessments of this loss focus on single metrics, such as species richness, and overlook the multidimensional effects of human activities. Because of its importance to ecosystem functioning, we investigated the functional diversity of medium- and large-sized mammal assemblages in protected and unprotected mountain forests in the Hengduan Mountains of southwest China. We hypothesized that anthropogenic pressures outside protected areas (PAs) favor generalist species and lead to functionally homogenized assemblages compared with assemblages inside PAs. Using generalized dissimilarity models, we also examined the key drivers shaping patterns of functional β diversity in these landscapes. Outside PAs, functional richness was markedly lower, trait space was significantly more homogeneous, and nestedness β diversity was more prevalent (57.6%) than inside PAs, clear indications of functional homogenization. Anthropogenic variables consistently outweighed environmental gradients and spatial distance in explaining variation in functional β diversity. Our results demonstrate that anthropogenic pressure affects trait composition in ways that species counts alone do not fully capture, emphasizing the importance of conservation strategies that protect both functional integrity and taxonomic diversity.

KEYWORDS

anthropogenic activity, beta diversity, camera trap, functional diversity, mountain forest, southwest China

INTRODUCTION

Human activities drive rapid defaunation, resulting in extinction rates far exceeding pre-human levels (Ceballos et al., 2015; Dirzo et al., 2014; Matthews et al., 2024). Besides species loss, these impacts profoundly affect other dimensions of biodiversity, including phylogenetic and functional diversity (Martínez-Núñez et al., 2023; Matthews et al., 2024). Over the past 50 years, a growing body of evidence has documented sharp declines across most biodiversity indicators (Dirzo et al., 2014; Martínez-Núñez et al., 2023; Tucker et al., 2018),

alongside accelerated biotic homogenization and widespread disruption of ecosystem functioning (Newbold et al., 2018; Su et al., 2021). Despite this, the consequences of human pressures on different facets of biodiversity remain poorly understood, limiting the ability to develop effective conservation strategies (Matthews et al., 2024; Toussaint et al., 2024).

Understanding biodiversity patterns and the processes that shape them is essential for advancing the knowledge of the natural world and developing effective conservation strategies amid escalating threats (Mokany et al., 2020; Soares et al., 2024). To ensure conservation efforts adequately represent biodiversity (Díaz & Malhi, 2022), scientists and policy makers must understand how diversity is distributed across space and time (Ferrier & Drielsma, 2010). However, most current biodiversity

Xueyou Li and Wenqiang Hu contributed equally to this work and are co-first authors.

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assessments still focus on taxonomic diversity, such as species richness, which does not account for the unique functional roles of species within ecosystems (Dornelas et al., 2014; Newbold et al., 2014; Outhwaite et al., 2020). It is increasingly recognized that not all species contribute equally to ecosystem functioning, and relying solely on taxonomic diversity provides an incomplete picture of biodiversity's ecological significance (Soares et al., 2024; Vernham et al., 2023; Yan et al., 2023). Although species richness is a widely used metric for exploring biodiversity patterns, it provides limited insight into the underlying mechanisms of community assembly (van der Sande et al., 2024) and often underrepresents other critical facets of biodiversity, such as functional and phylogenetic diversity (Brun et al., 2019; Cadotte & Tucker, 2018).

The ecological roles of species are shaped by their functional traits—the ecological and morphological characteristics that influence an organism's performance and fitness (Matthews et al., 2024; Tobias et al., 2022). Functional traits are fundamental for understanding and predicting community assembly processes (Latombe et al., 2024) and the effects of global change drivers on ecosystem functioning and resilience (Boinot & Alignier, 2023). Over the past 2 decades, the use of functional diversity—the range of functional traits in an assemblage—has gained traction as a powerful tool for exploring the mechanistic impacts of extinctions on ecosystem functioning, offering insights beyond those provided by taxonomic methods (Ali et al., 2023; Carmona et al., 2021). Functional diversity is increasingly recognized as a key predictor of community assembly processes (Petchey & Gaston, 2006; Schleuter et al., 2010) because it reveals ecological gradients and highlights the impacts of environmental threats on ecosystem functioning (Graco-Roza et al., 2022; Pakeman, 2011). This makes it a more effective index for predicting the functional consequences of changes in anthropogenic activity on biodiversity (Eastwood et al., 2023; Rodríguez-Caro et al., 2023). Understanding how global change drivers influence functional diversity and functional vulnerability has become a central goal for ecologists (Auber et al., 2022; Cooke, Eigenbrod, et al., 2019; Loiseau et al., 2020). Theoretical frameworks suggest that negative deviations in functional biodiversity measures indicate habitat filtering, whereas positive deviations point to neutral assembly processes (Mouchet et al., 2010). However, despite the growing availability of functional diversity estimates, they are still rarely applied to field survey data spanning environmental and anthropogenic disturbance gradients. This gap emphasizes the need for further empirical studies that test and refine the predictive power of these metrics under real-world conditions.

Changes in the composition of biotic assemblages along environmental gradients are critical for understanding the ecological processes that structure and sustain biodiversity patterns (García-Navas et al., 2022; Graco-Roza et al., 2022). This site-to-site turnover in assemblage composition, known as beta diversity, provides valuable insights into the spatial dynamics of biodiversity (White et al., 2024). Beta diversity has been extended to encompass multiple facets of diversity, including not just species composition but also functional diversity, capturing how assemblages differ in their ecological functions

(García-Navas et al., 2022). By estimating the degree of compositional dissimilarity among sites (Baselga, 2013), beta diversity can quantify ecological uniqueness, often through pairwise comparisons between focal sites and others (Mokany, Ware, et al., 2022). A decline in beta diversity signals biotic homogenization, a significant and well-documented threat to global biodiversity, ecosystem functioning, and resilience (Boinot & Alignier, 2023). As a result, beta diversity has become a key metric for assessing the effectiveness of conservation strategies and identifying priority habitats for protection. By highlighting areas of ecological distinctiveness or vulnerability, beta diversity serves as an essential tool for guiding conservation planning and mitigating the impacts of anthropogenic change (Mokany et al., 2020).

Deciphering the multifaceted patterns of biodiversity and the ecological drivers shaping them across environmental gradients provides critical insights into the conservation value and resilience of communities (Rolls et al., 2023; Tsang et al., 2023). Niche-based theory posits that deterministic processes, such as abiotic and biotic filtering, play a dominant role in structuring biodiversity patterns (Chave, 2004). In contrast, neutral theory suggests that species are ecologically equivalent and that stochastic processes primarily govern their distributions (Vellend et al., 2014). Mountain forests offer exceptional opportunities to investigate these dynamics because they exhibit rapid changes in community composition and abiotic conditions over short distances (Rahbek et al., 2019). Mountain regions are biodiversity hotspots and contribute disproportionately to global terrestrial biodiversity, hosting over 85% of the world's amphibian, bird, and mammal species (Rahbek et al., 2019). However, these ecosystems are particularly fragile and face escalating human disturbances, including settlement expansion, wood harvesting, and pastoral activities (Li, Hu, Bleisch, Li, Wang, Ti, et al., 2022). Increasing human activity in mountain regions continues to drive habitat conversion, creating mosaics of fragmented landscapes and threatening the ecological integrity of these vital ecosystems (Perez-Quezada et al., 2023). In addition, climate change exacerbates human-induced pressures in mountain systems, driving elevational range shifts that heighten extinction risk, particularly for high-elevation endemics, and interacts synergistically with land-use change threats (Bühne et al., 2021; Dirnböck et al., 2011; Elsen et al., 2020; Pepin et al., 2015).

Human modification (HM) and human presence (HP) in mountain forests pose significant threats to wildlife, particularly medium- and large-sized mammals, which are highly susceptible to anthropogenic activities (Li, Hu, Bleisch, Li, Wang, Lu, et al., 2022). These mammals play crucial roles in maintaining mountain forest ecosystems by contributing to seed dispersal, nutrient cycling, and vegetation recruitment (Lacher et al., 2019). Their diverse assemblage spans a wide range of dietary niches, home range sizes, and body sizes, enabling them to perform varied and essential functions in ecosystem dynamics (Gorczynski et al., 2023). However, many medium- and large-sized mammals are endangered, putting the critical ecosystem services they provide at risk (Toussaint et al., 2021). Given the accelerating biodiversity loss of the Anthropocene, understanding the specific aspects of human impacts on medium- and large-sized mam-

mal diversity is increasingly urgent, particularly for functional diversity, because it links species' ecological roles to their environment at local scales, offering deeper insights into ecosystem resilience and functioning (Petchey & Gaston, 2006). Despite the critical role of human impacts on biodiversity loss, many analyses still focus solely on single metrics like species richness, neglecting the multidimensional effects of human activities (Li, Hu, Bleisch, Li, Wang, Ti, et al., 2022; Polaina et al., 2018; Semper-Pascual et al., 2023). Furthermore, field-based studies examining the relationship between human impacts and mammal functional diversity remain scarce, leaving significant gaps in understanding of how these assemblages respond to varying levels of human pressure.

We used camera-trap data to estimate the functional diversity of medium- and large-sized mammals in the Hengduan Mountains of southwest China, a global biodiversity hotspot, to assess the functional diversity patterns inside and outside protected areas (PAs). Specifically, we examined changes in alpha and beta functional diversity, defined as the hypervolume of functional space occupied by mammal assemblages. We investigated whether change patterns are influenced by site conservation status (protected vs. unprotected, reflecting resource availability and shelter conditions) and anthropogenic factors (HP and habitat modification).

Given the well-documented threats human activities pose to mountain biodiversity, we expected human impacts to play a significant role in shaping biotic changes across mountain forests (He et al., 2023; Li, Hu, Bleisch, Li, Wang, Lu, et al., 2022). Specifically, at the alpha diversity level, we hypothesized that assemblages outside PAs exhibit lower functional richness than species richness (functional underdispersion) due to habitat fragmentation. Human activities can shift mammal communities toward relatively small, generalist species (Haight et al., 2023; Li, Hu, Bleisch, Li, Wang, Ti, et al., 2022). Consequently, at the level of beta diversity, we expected assemblages outside PAs to be more functionally homogeneous, reflecting the loss of species with unique ecological roles (Clavel et al., 2011) and resulting in reduced functional beta diversity compared with those inside PAs. Previous research in this region highlighted disproportionate losses of threatened terrestrial mammals along gradients of human disturbance (Li, Hu, Bleisch, Li, Wang, Ti, et al., 2022). However, the implications of these losses for functional diversity remain poorly understood. We sought to provide new insights into the drivers of functional diversity patterns in medium- and large-sized mammals across mountain forests and inform targeted conservation strategies.

METHODS

Study area

We conducted this study in the Gaoligong Mountains (GLG) and Biluo Snow Mountains (BL) in southwestern China (Figure 1). The 2 mountain ranges are parallel to each other and are part of the Mountains of Southwest China, a global biodiversity hotspot known for its rich and unique flora and fauna

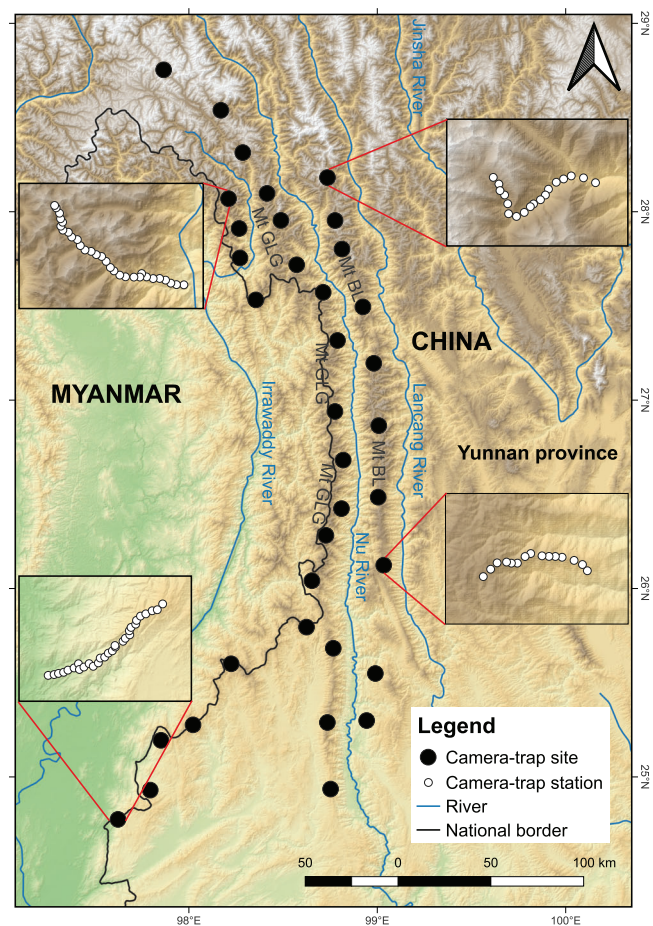


FIGURE 1 Location of the 2 study areas in southwestern China: the Gaoligong Mountains (GLG) and the Biluo Snow Mountains (BL).

(CEPF, 2022). Despite their similar climatic and topographic conditions, the 2 ranges differ significantly in conservation status and human pressure, providing an ideal system to examine functional diversity patterns inside and outside PAs. The GLG is strictly protected, with minimal human disturbance and intact habitats, whereas the BL is unprotected and subject to intense human activities (Li, Hu, Bleisch, Li, Wang, Lu, et al., 2022).

The region's climate is influenced by the southwest monsoon originating from the Indian Ocean. Its dramatic topography features numerous peaks exceeding 4000 m in elevation. Along the elevational gradient, vegetation types transition through distinct zones, including tropical monsoon forest (<1000 m), subtropical evergreen broad-leaf forest (1000–2600 m), temperate deciduous broad-leaf forest (1000–3000 m), subalpine coniferous forest (2700–3500 m), and alpine meadow (>3400 m) (Wang et al., 2007). The region is also home to several of the world's most threatened mammals, such as the clouded leopard (*Neofelis nebulosa*), dhole (*Cuon alpinus*), and takin (*Budorcas taxicolor*) (Li, Li, et al., 2024). HM across the region varies and includes changes associated with livestock grazing, natural resource extraction, and transportation (Li, Hu, Bleisch, Li, Wang, Lu, et al., 2022). These anthropogenic pressures, combined with the region's exceptional biodiversity and contrasting

conservation contexts, make it a valuable system for studying the impacts of human activities on functional diversity.

Camera trapping data

As part of a long-term monitoring program in the Hengduan Mountains, a component of the China Biodiversity Monitoring and Research Network (Sino BON), we deployed 836 camera traps across mountain forests with a minimum spacing of 1 km between cameras. We placed 695 cameras in the GLG (PA) and 141 in the BL (unprotected area). Camera traps were installed approximately 1 m above ground and positioned away from trails frequently used by humans to minimize the risk of vandalism. Each camera remained active at its site for a full year. Batteries and SD (secure digital) cards were replaced every 4 months.

Sensitivity was set to low to reduce false triggers from nonanimal movements. The cameras were programmed to capture 3 photos per detection event, with a minimum delay of 3 s between subsequent detections. Photographs were identified to the species level whenever possible (see Appendix S1 for some detected wildlife photographs). HP (including records of people, livestock, and dogs) was documented to estimate the intensity of human activity at each site. All images of the same species (including humans) from the same camera were treated as independent records if separated by at least 1 h. Data collection occurred from October 2020 to September 2022. Despite some camera stations failing to provide usable data due to malfunctions or loss, the final numbers of operational traps contributing data were 631 in GLG and 110 in BL.

Anthropogenic and habitat covariates

Satellite-derived HM data were obtained from the Global Human Modification map at a 1-km resolution (Kennedy et al., 2019). Patch density was extracted from the ESA WorldCover map (<https://viewer.esa-worldcover.org>). To quantify HP at each camera-trapping station, we calculated relative abundance indices (RAIs) based on independent detections of domestic dogs, people, and livestock, expressed as detections per 100 camera-trap days. We also measured habitat covariates that potentially influence species ecological performance and the functional diversity of mammal assemblages (Li, Hu, Bleisch, Li, Wang, Lu, et al., 2022). These included annual mean temperature, temperature seasonality, annual precipitation, precipitation seasonality, percent forest cover, net primary production (NPP), and elevation.

Climate variables were obtained from the WorldClim database (<http://www.worldclim.org/bioclim>). Percent forest cover was derived from 250-m resolution Moderate Resolution Imaging Spectroradiometer (MODIS) imagery (MOD44B Vegetation Continuous Fields product) for 2022 (<https://ladsweb.modaps.eosdis.nasa.gov/missions-and-measurements/products/MOD44B>). The NPP values for each camera-trapping site were sourced from MODIS satellite

images for the same year (<https://doi.org/10.5067/MODIS/MOD17A3H.006>). For each camera-trapping station, raster-based covariates (HM, NPP, percent forest cover, and climate variables) were extracted from within a 1-km radius around the sampling station. To address the skewed distribution of HP data, we applied a log transformation ($\log[x + 0.1]$). Other continuous variables were standardized (mean = 0, SD = 1) to homogenize distributions and minimize the influence of outliers.

We tested for collinearity among covariates with a threshold value of $|r| = 0.7$ (Dormann et al., 2013). Based on this test, we removed annual mean temperature and annual precipitation to avoid multicollinearity because they were highly correlated with NPP ($r = 0.82$) and elevation ($r = 0.79$), respectively. We retained NPP and elevation for 2 reasons: ecological relevance and model performance. Productivity is a direct predictor of resource availability for medium- and large-sized mammals (Li et al., 2021), and elevation integrates multiple fine-scale habitat filters that strongly influence community composition (Jacquier et al., 2020). Preliminary variance inflation factor (VIF) tests indicated no significant multicollinearity ($VIF < 4$ for all retained variables). Models retaining NPP and elevation explained more deviance than those retaining temperature and precipitation.

Species trait data

We focused on medium- and large-bodied terrestrial mammals with body mass >500 g because these species exhibit large spatial and dietary requirements, high behavioral plasticity, and sensitivity to human activities (Li, Hu, Bleisch, Li, Wang, Lu, et al., 2022). Mammal trait data were compiled from a global mammal trait database (Cooke, Bates, et al., 2019) and a database specific to Chinese mammals (Ding et al., 2022). For each species, we collected data on 4 functional traits—body mass, home range, diet, and activity period—that are commonly used to evaluate mammal functional diversity (Gorczynski et al., 2023). These traits broadly reflect the quantitative and qualitative environmental requirements of species and are effective in capturing their responses to environmental changes (Gorczynski et al., 2023). For species with missing trait data, we imputed mean values for all congeneric species in our dataset. This approach leverages phylogenetic niche conservatism; closely related species tend to share similar functional traits (Wiens et al., 2010). Body mass and home range data were log-transformed to ensure comparability across traits with different value ranges.

Data analyses

We used coverage-based rarefaction and extrapolation curves to inspect our sampling completeness. For each site (i.e., BL and GLG), rarefaction curves were produced using 1000 randomizations (Hsieh et al., 2016). The species accumulation curves for both areas showed a clear saturation trend, indicating that our sampling efforts were sufficient to capture the majority of

the medium- and large-sized mammal species present in the study areas (Appendix S1).

We evaluated functional diversity with hyperdimensional functional trait spaces derived from probabilistic hypervolumes (Mammola & Cardoso, 2020). Probabilistic hypervolumes accounted for the fact that not all areas within the boundaries of a trait space are occupied with equal intensity. They allow the identification of regions with lower or higher density in the trait space, effectively reflecting the uneven probabilities of encountering species with specific trait combinations (Mammola & Cardoso, 2020). Functional trait hypervolumes require a minimum number of species to estimate bandwidths reliably, and communities with <3 species often yield singular trait profiles and unstable volumes (Mammola & Cardoso, 2020). Following the guidance of Mammola et al. (2024), we therefore excluded such cases, resulting in 488 usable camera traps (434 in GLG and 54 in BL). To construct the global trait space, we calculated pairwise functional distances among all medium- and large-sized mammal species based on their traits. We used Gower's distance to calculate pairwise distances because it is suitable for mixed data types (e.g., numeric and categorical) and appropriately standardizes them (Coutant et al., 2023). The resulting dissimilarity matrix was then projected into a multi-dimensional space using principal coordinate analyses (PCoA), with Cailliez correction to address potential negative eigenvalues (Soares et al., 2024). To represent the global trait space, we retained the first 4 PCoA axes, which accounted for 91% of the total variation.

Using the global trait space, we constructed hypervolumes employing a Gaussian kernel density estimator with a default bandwidth for each axis (Blonder et al., 2018). Abundance-based measures of beta diversity inherently assign less weight to rare species (Socolar et al., 2016). However, rare species often support more vulnerable functions and disproportionately enhance the potential breadth of functional space within communities, thereby sustaining local ecosystem processes (Mouillot et al., 2013; Toussaint et al., 2021). Thus, we transformed the independent records for each species across sites into a presence-absence matrix, with rows representing camera-trapping sites and columns representing individual species. We then calculated a set of functional diversity metrics with hypervolume-based functions (Mammola & Cardoso, 2020) in the R package BAT (Cardoso et al., 2021). We measured the functional richness of each assemblage (alpha diversity) as the total volume of each hypervolume. We also measured the evenness of each hypervolume, which represents the regularity of the distribution of functional elements in the total trait space (Mammola & Cardoso, 2020). Furthermore, we calculated the relative abundance-weighted functional diversity (independent detections per 100 camera-trapping days) to explore additional functional differences between mammal assemblages inside PAs versus outside PAs. Functional richness and evenness derived from the binary and relative abundance-weighted approaches were similar inside and outside PAs (Appendix S1).

Because our primary focus was the potential functional space occupied by the assemblage, rather than the magnitude of functional processes driven by species abundance, we conducted further analyses of functional diversity based on

presence-absence data. We performed Shapiro-Wilk tests for normality on the functional diversity metrics (richness, evenness, β diversity). Only functional β diversity data met the normality assumption (Shapiro-Wilk $W = 0.964$, $p = 0.110$), whereas both functional richness ($W = 0.992$, $p = 0.007$) and evenness ($W = 0.898$, $p < 0.001$) significantly deviated from normality. Consequently, we used independent sample t tests to compare functional β diversity between BL and GLG and Mann-Whitney U tests to compare functional richness and evenness.

To address the imbalance in sampling effort inside versus outside PAs, we randomly selected 54 stations from GLG 99 times and calculated functional richness and functional evenness for each sample. We also calculated functional richness and functional evenness based on the complete camera-trapping data from GLG. We then used one-sample t tests to determine whether the mean functional richness and functional evenness from random samples differed significantly from the estimates based on the full dataset. Functional richness and evenness derived from subsamples did not differ from full-data estimates (functional richness: $t = 0.273$, $p = 0.785$; functional evenness: $t = 0.932$, $p = 0.354$). Nonetheless, to avoid potential bias introduced by arbitrarily selecting a smaller subset from GLG and ensure that functional diversity metrics were based on the full sampling effort in each region, we based all analyses on the complete datasets for both mountains.

To quantify changes in functional diversity across sites, we calculated pairwise functional β diversity among camera-trapping sites with the Sørensen dissimilarity index (Anderson et al., 2011). This beta diversity metric was further decomposed into 2 ecological processes underlying overall dissimilarity (β total) among kernel hypervolumes: replacement of trait space between hypervolumes (β replacement) and loss or gain of space enclosed by each hypervolume (β richness) (Legendre, 2014). Each beta component ranges from 0 to 1, with 0 indicating no change in functional composition and 1 indicating no overlap between trait spaces (Mammola & Cardoso, 2020).

To determine how unusual the observed values of functional richness were and to control for the effect of species richness, we calculated the standardized effect size (SES) of functional richness. For each camera-trapping site, we used a null model with 999 iterations per assemblage in the permatasp (quasiswap) algorithm from the R package vegan. This algorithm preserves the matrix fill and the row and column sums of the raw camera-trapping data for occurrences of medium- and large-sized mammals in the randomized matrices. The randomization was stratified by conservation status (PAs and unprotected area). That is, the null model was based on the assumption that exchangeability was possible only in areas with a similar conservation status and did not incorporate taxa absent from the focal conservation types. Positive SES values indicated functional overdispersion (greater trait diversity than expected), whereas negative SES values indicated functional underdispersion (less trait diversity than expected). We assessed significant departures of the observed values from the null distribution by ranking them within the null distribution and dividing by the total number of simulated values to obtain quantile scores. Specifically, observed values were deemed significant if they

ranked in the lower 2.5% or upper 2.5% of the null distribution (Mammola et al., 2024).

To assess in detail the influence of individual environmental and anthropogenic variables on the functional beta diversity of medium- and large-sized mammal assemblages, we employed spatially explicit generalized dissimilarity modeling (GDM) (Mokany, Ware, et al., 2022). A GDM is an extension of matrix regression techniques that allows incorporation of spatial and environmental distances to predict β diversity (Mokany, Ware, et al., 2022). It models the relationship between β diversity and spatial, environmental, or both types of distances with I spline basis functions in a generalized linear modeling framework, which allows for nonlinear relationships between dissimilarity and distance (Fitzpatrick et al., 2013). A major advantage of GDM over traditional modeling approaches, such as the Mantel test for correlation between distance matrices, is its ability to explicitly and simultaneously account for the influence of both spatial distance and environmental variables when explaining biotic variation (Mokany, McCarthy, et al., 2022). We fitted the models with the R package *gdm* with the default setting of 3 I spline basis functions per predictor (Mokany, Ware, et al., 2022). We plotted the partial effect of each predictor on functional beta diversity while holding all other predictors constant. Additionally, we summed the I spline coefficients to assess the relative importance of each variable in shaping patterns of beta diversity (model code in Appendix S2).

RESULTS

Forty-eight medium- and large-sized mammal species were detected (36 in BL and 47 in GLG), of which 23 species were herbivores, 17 were carnivores, and 8 species omnivores (Appendix S3). Species ranged in average body mass from 0.5 kg (*Mustela kathiah*) to 302 kg (*B. taxicolor*). At camera-trapping sites, the HM index ranged from 0.02 to 0.62, and the RAIs of instant HP ranged from 0 to 118.66. Elevation ranged from 604 to 4306 m. Mean functional richness ranged from 0.001 to 0.298, and functional evenness ranged from 0.146 to 0.849.

Patterns of functional α diversity

Functional richness was significantly lower in unprotected forests in BL than in protected forests in GLG ($Z = -5.502$, $p < 0.001$) (Figure 2a), suggesting the dominance of species with specific functional traits outside PAs. Traits of medium- and large-sized mammal assemblages in unprotected forests were significantly more homogeneous than those in protected forests (evenness: $Z = 3.729$, $p < 0.001$) (Figure 2b; Appendix S1).

Values of SES for the functional richness of each assemblage in unprotected forests were left skewed, with 70.4% of sites displaying a prevalence of functional underdispersion (SES values below 0) over functional overdispersion (Figure 2c). Still, most of these SES values were closer to zero (Figure 2c). Only 5 sites (9.2%) had all SES values below 0 ($p < 0.05$). In protected forests, however, SES values for the functional richness of each assemblage were slightly right skewed, with 56.2% of sites show-

ing a prevalence of functional overdispersion (SES values above 0) over functional underdispersion (Figure 2c). Six assemblages (1.3%) had all SES values above 0 ($p < 0.05$).

Patterns of functional β diversity

Total functional β diversity (β total) was significantly higher in pairwise comparisons between camera-trap sites in unprotected forests. In contrast, assemblages in protected forests were, in general, more similar to one another ($t = 47.284$, $df = 1482$, $p < 0.001$) (Figure 3a). Overall, most of the variation in β diversity was due to nestedness of trait space among assemblages, especially across assemblages outside PAs (Figure 3c). The β richness accounted for 57.6% and 53.2% of the variation in β diversity in BL and GLG, respectively (Figure 3b). Both β replacement and β richness were significantly higher in BL compared to GLG (β replacement: $t = 10.585$, $df = 1452$, $p < 0.001$; β richness: $t = 19.590$, $df = 1452$, $p < 0.001$), with patterns largely mimicking the differences in total β diversity (Figure 3).

Drivers of functional change

Results from the GDMs showed that anthropogenic and environmental variables, rather than geographic distance, influenced the observed total β diversity patterns in medium- and large-sized mammal assemblages of the 2 mountains (Figure 4a). The relative importance of variables differed between protected and unprotected mountain forests (Figure 4a). The most important variables explaining total functional dissimilarity patterns were HP for BL (52.3%) and HM for GLG (40.8%) (Figure 4b). The contribution of geographic distance to total functional β diversity was negligible in both mountains (below 0.001). The deviance explained by the fitted GDM was 17.8% for BL and 5.49% for GLG.

The I splines (response curves) fitted to the explanatory variables in the GDM models showed that functional dissimilarity was strongly associated with changes in HP in BL (Figure 4c). Changes in functional diversity across assemblages in BL appeared to stabilize above a threshold value of temperature seasonality and elevation (Figure 4e,f). Functional β diversity increased asymptotically with both HM and elevation in GLG (Figure 4d,f).

The relative importance of environmental predictors differed between the 2 β diversity components (Figure 5). The most important variables explaining the β replacement were HP for BL (34.0%) and HM for GLG (42.5%) (Figure 5a). The most important variables explaining the β richness were HP for BL (71.6%) and temperature seasonality (BIO4) for GLG (36.9%) (Figure 5b).

DISCUSSION

Biodiversity loss in the Anthropocene is eroding functional diversity, a cornerstone of ecosystem services (Rodríguez-Caro et al., 2023). PAs remain the backbone of global conserva-

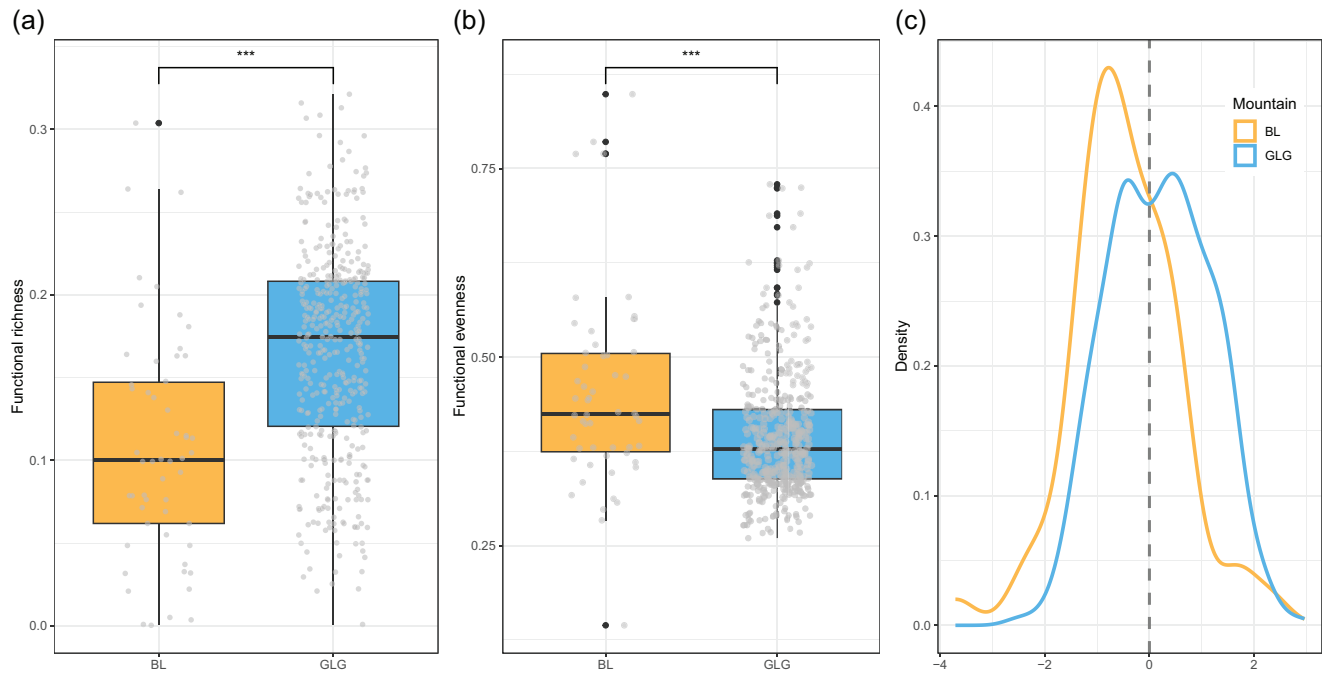


FIGURE 2 The α -level functional diversity of medium- and large-sized mammal assemblages in mountain forests of southwestern China: (a) functional richness (FRic) and (b) functional evenness at camera-trap sites in protected forests (GLG) versus unprotected forests (BL) (horizontal lines, median; box ends, interquartile range; whiskers, values within 1.5 times the interquartile range; black dots, outliers beyond the whiskers; gray dots, raw site values; asterisks, significant differences between land-use types [*** $p < 0.001$]) and (c) kernel-density distribution of standardized effect sizes (SEs) for FRic across sites in protected and unprotected forests (vertical dashed line, SES = 0, i.e., no deviation from the null expectation).

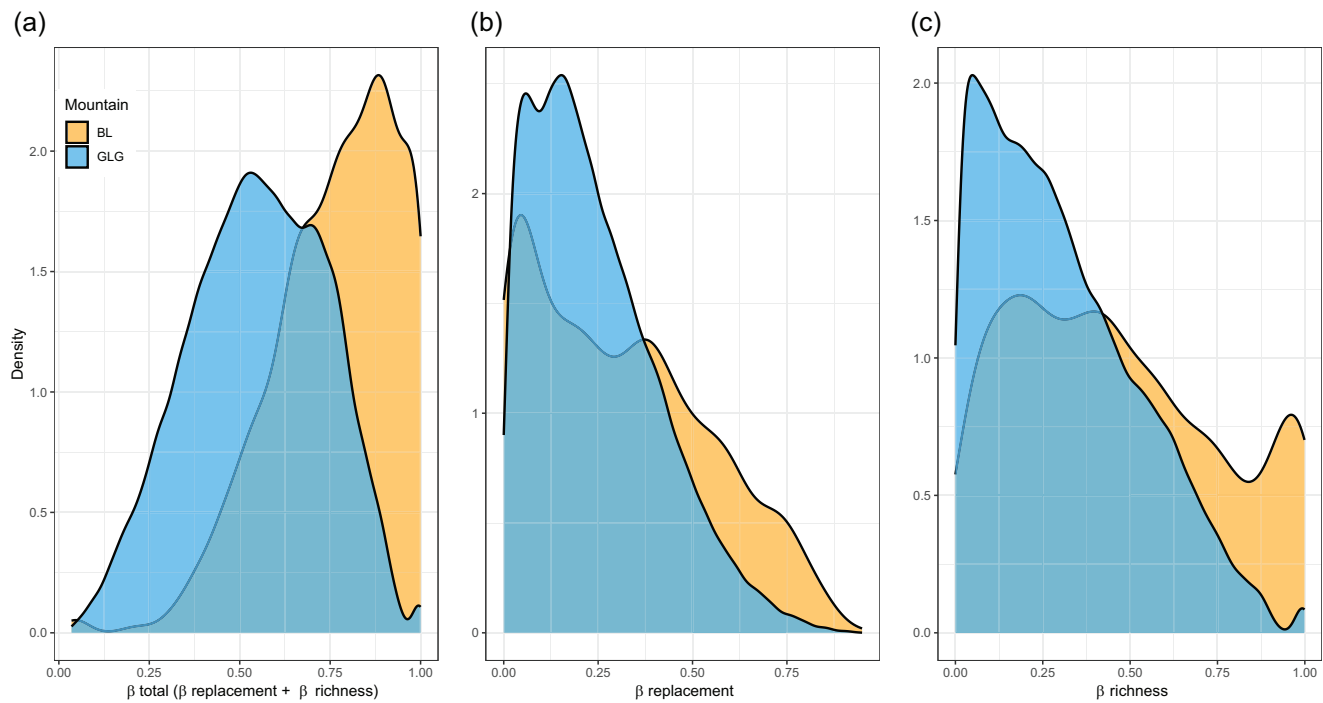


FIGURE 3 Density distributions of pairwise functional β diversity among medium- and large-sized mammal assemblages in protected (GLG) and unprotected (BL) mountain forests of southwestern China: (a) total functional dissimilarity between communities, (b) differences in functional composition due to replacement of functional space, and (c) differences in the amount of functional space enclosed in each hypervolume.

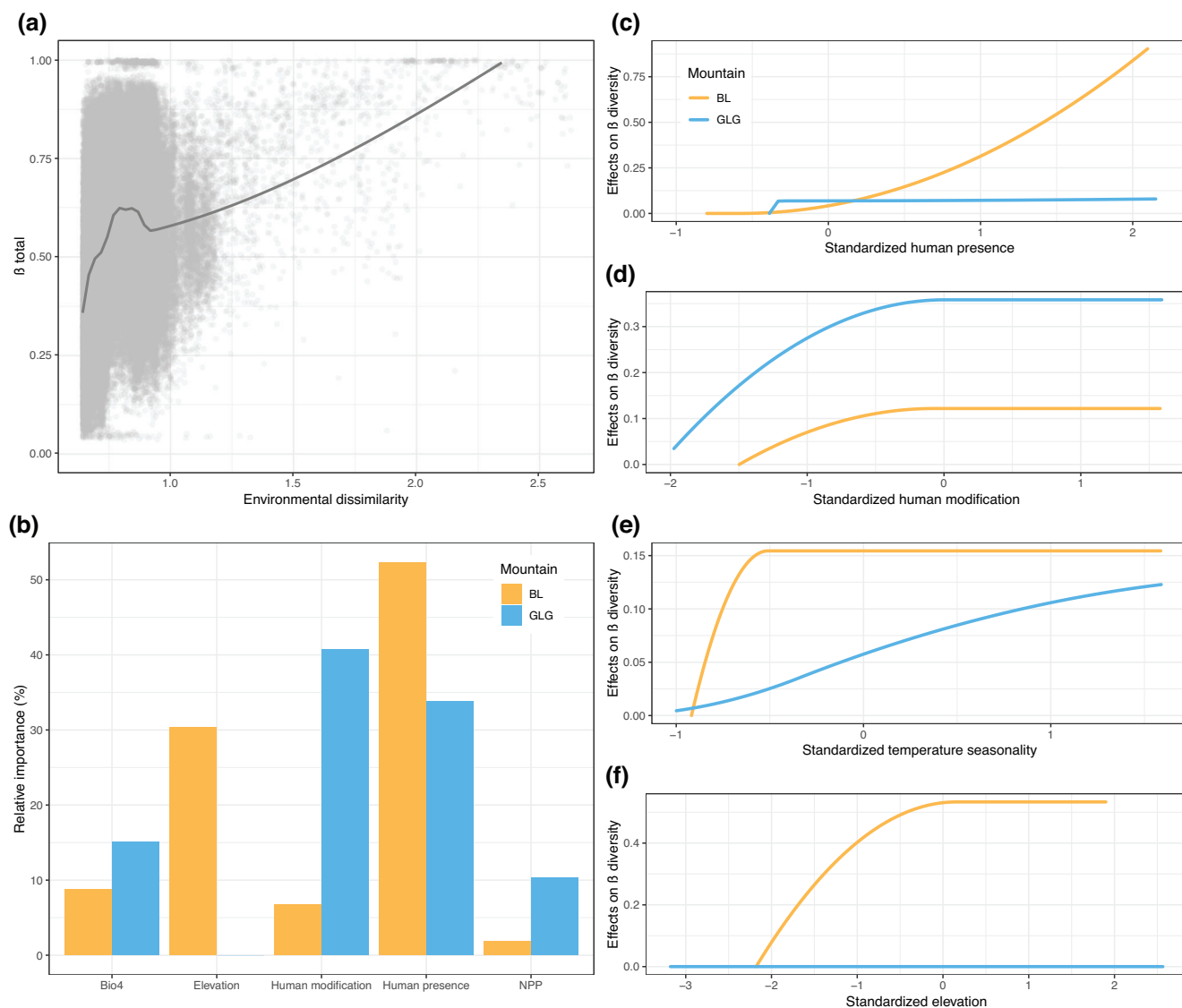


FIGURE 4 Generalized dissimilarity model results for camera-trap presence and absence data on medium- and large-sized mammals in protected (GLG) and unprotected (BL) mountain forests in southwestern China: (a) fitted relationships between total functional β diversity (β total) and environmental dissimilarities in human pressure and human modification, net primary productivity (NPP), temperature seasonality (Bio4), and elevation, (b) relative contribution (percent deviance explained) of each predictor to β total, and (c–f) partial ecological distance plots showing the independent effects of each variable on β total for protected (blue) and unprotected (yellow) sites (curve shape indicates how the rate of change in functional trait space varies along each environmental gradient).

tion strategies to stem this decline (Brodie et al., 2023), and the Kunming–Montreal Global Biodiversity Framework now calls for safeguarding 30% of Earth’s land and sea by 2030 (“30 × 30”) (Sala et al., 2021). Demonstrating that legal protection itself improves multiple facets of biodiversity is therefore vital. Unprotected forests showed sharp declines in functional richness relative to species richness, pronounced functional clustering, and nestedness-dominated β diversity—indications of functional homogenization. In contrast, PAs maintained higher functional richness and turnover. Anthropogenic variables, rather than environmental distance, were the primary drivers of functional β diversity across all sites. Collectively, our results showed that PA designation substantially bolstered the functional diversity of medium- and large-sized mammals,

underscoring the pivotal role of PAs in meeting the 30 × 30 conservation target.

Drawing on in situ camera-trap surveys from 2 parallel, isolated mountain ranges in the Hengduan Mountains of southwest China, ranges that share comparable climate and terrain yet differ in their protection status, we found that PAs safeguarded functionally distinct medium- and large-sized mammals more effectively than unprotected sites (Greco et al., 2025). As habitat conversion and fragmentation escalate with mounting human pressures, the long-term persistence of most wildlife is likely to rely even more heavily on PAs (Williams et al., 2022). Previous assessments of PA performance have produced mixed results (Brodie et al., 2023; Stewart et al., 2019). Nevertheless, mountain forest PAs typically have lower levels of HM than matched

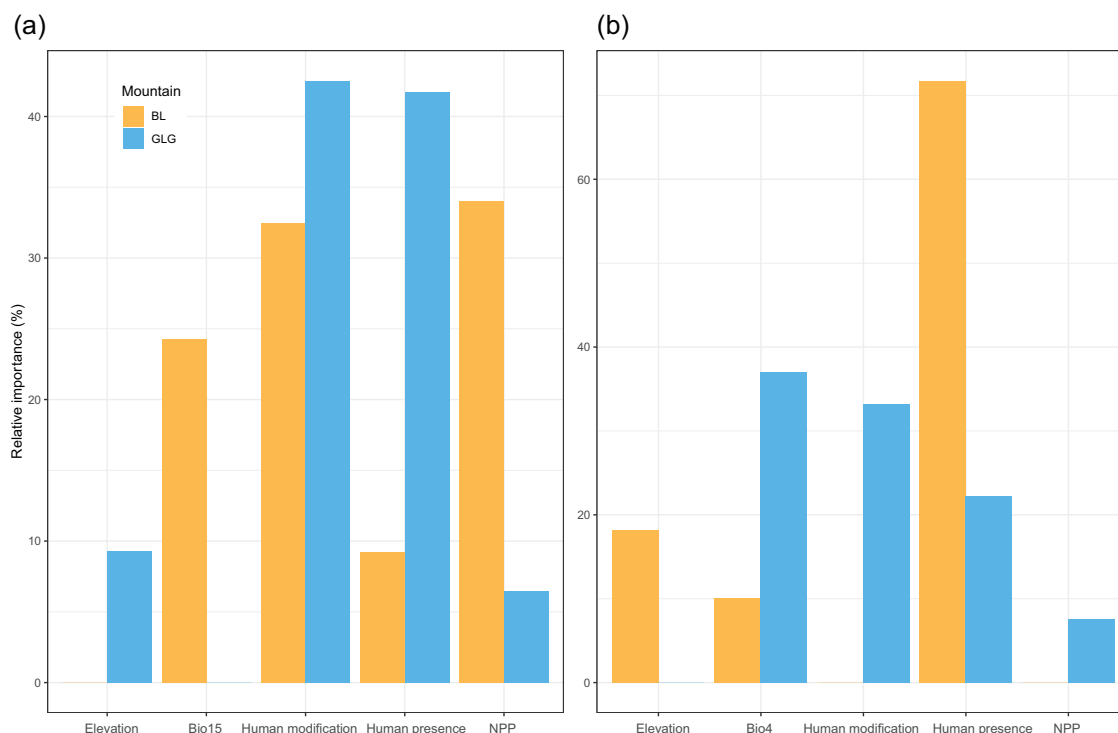


FIGURE 5 Relative contribution (% deviance explained) of each predictor to (a) β replacement (i.e., replacement of trait space between hypervolumes) and (b) β richness (i.e., loss or gain of space enclosed by each hypervolume) (NPP, net primary productivity).

unprotected areas (Geldmann et al., 2019) and often support higher biodiversity (Chen et al., 2022; Magioli et al., 2021; Pacifici et al., 2020). Designed chiefly to protect rare and threatened taxa, many PAs have indeed proved instrumental in species recovery (Chen et al., 2022; Fornitano et al., 2024; Jhala et al., 2025; Magoulick et al., 2024). Globally, threatened mammals are recovering faster inside PAs than outside them (Magoulick et al., 2024).

However, because PAs are often sited in remote regions with inherently low human pressure to minimize opportunity costs (Venter et al., 2016), their apparent benefits can be confounded by preexisting ecological conditions (Brodie et al., 2023; Chen et al., 2022). Biodiversity inside some PAs may have been high—or already increasing—before designation, and positive trends can mask ongoing declines in absolute diversity (Magoulick et al., 2024). Indeed, PA status alone seldom predicts mammal diversity across spatial scales (Stewart et al., 2019). Conservation outcomes also depend on natural features and human activities in the surrounding landscape (Fornitano et al., 2024; Jones et al., 2018; Stewart et al., 2019). Our results reinforce this point: human pressures are the dominant drivers of functional β diversity. This underscores the critical role of effective PA management in maintaining mammalian functional diversity in increasingly human-modified environments.

Most broadscale biodiversity assessments still focus on taxonomic metrics, such as species richness (Magioli et al., 2021) or population density (Greco et al., 2025). Yet, anthropogenic pressures frequently trigger cascading losses in the functional diversity that underpins ecosystem processes (Brodie et al., 2021). By comparing protected and unprotected sites,

we provided the missing functional perspective for Hengduan Mountain mammals. Our results showed that PAs are indispensable for preserving mammalian functional diversity and the ecological roles these species perform, highlighting their irreplaceable value in an increasingly human-modified landscape.

We compared functional diversity metrics between 2 mountain forests with similar climatic and topographic conditions but contrasting conservation designations, providing insight into the efficacy of PAs in protecting functional diversity. As predicted, our results indicated that the functional richness of medium- and large-sized mammal assemblages outside PAs was predominantly underdispersed (Figure 2). Striking declines in functional diversity may, in turn, affect ecosystem functioning (Gorczynski et al., 2021).

Previous studies on global and regional variation in functional diversity of mammal and bird assemblages have identified anthropogenic disturbance as a major driver of functional diversity loss (Flynn et al., 2009; Gorczynski et al., 2021; Newbold et al., 2014). Mountain forests in southwest China are fragile and subject to anthropogenic drivers of degradation, including transportation, resource collection, and livestock grazing, particularly outside PAs (Li, Hu, Bleisch, Li, Wang, Lu, et al., 2022). Compared with mammal assemblages in PAs, functional richness outside PAs was significantly lower, indicating the loss of functional traits in unprotected habitats. Functional evenness, representing the regularity of species relative abundance distribution in functional spaces (Mouchet et al., 2010), was significantly higher in unprotected areas than in PAs. This suggests that mammal assemblages were more homogeneous

regarding functional groups in the unprotected areas. Zhao et al. (2024) also found significantly higher functional evenness in fish communities in unprotected areas. This could be due to the loss of rare and specialist species with unique functional spaces in fragmented habitats (Filippi-Codaccioni et al., 2009). Lower evenness in PAs indicated that these habitats supported more rare species with unique trait spaces, which skewed species occurrence across camera-trapping sites and decreased the evenness facet of functional diversity.

Our presence-absence-based approach primarily reflects the breadth of functional trait combinations in mammal assemblages at each camera-trapping site. Although this approach is valuable for assessing the vulnerability of functional space (especially with rare species carrying unique traits), it might underrepresent the functional dominance of common species.

Our results showed that among-site variation in functional diversity across unprotected mountain forests was predominantly driven by differences in richness (Figure 3). This dominance is expected when species susceptible to the loss of certain environmental conditions or habitats are selectively removed from homogeneous habitats (García-Navas et al., 2025).

Landscape patterns of functional β diversity have significant implications for the single large reserve or several small reserves debate (Simberloff & Abele, 1976). In landscapes with large differences in richness, conservation efforts could better focus on the richest sites at the expense of low-diversity areas (Socolar et al., 2016). Thus, our findings highlight the importance of considering the functional dimension of biodiversity to implement targeted and more effective conservation practices (Rodríguez-Caro et al., 2023).

Contrary to our expectations, functional dissimilarity among medium- and large-sized mammal assemblages across unprotected areas was significantly higher than in PAs (Figure 3). Why do more disturbed and fragmented environments exhibit a higher rate of functional composition change (i.e., higher beta diversity)? This could be explained by stronger stochastic processes under low habitat suitability and more stressful conditions (García-Navas et al., 2021). In human-modified landscapes, food resources and habitats are often fragmented and unevenly distributed, increasing stochastic replacement and nestedness in the functional trait spaces of wildlife species (García-Navas et al., 2025). Previous studies have reported a marked decline of threatened mammal species associated with human-modified landscapes (Li, Hu, Bleisch, Li, Wang, Ti, et al., 2022), which can have knock-on effects (i.e., indirect or cascading impacts throughout an ecosystem) on functional composition at the assemblage level. For instance, the decline of large predators can alter intraguild interactions and trigger mesopredator outbreaks, in turn leading to increased killing of smaller prey (Ritchie & Johnson, 2009). Human-induced changes in functional composition across unprotected landscapes might erode ecosystem functioning, resistance, and resilience (Boinot & Alignier, 2023). Based on our results, we conclude that protected landscapes exhibit lower spatial heterogeneity in functional composition than those in more disturbed and less favorable habitats.

An improved understanding of the relationships between the functional diversity of mammal assemblages and environmental gradients is crucial, given the accelerating rates of land-cover change and disturbance globally (Schumm et al., 2019). Information on the responses of mammal functional diversity to human activities at management-relevant scales is currently scarce. As predicted, anthropogenic variables, rather than geographic distance, were the most important predictors shaping the observed β diversity patterns for medium- and large-sized mammal assemblages in both protected and non-protected landscapes (Figure 4). The relative importance of environmental predictors differed between functional β diversity components, revealing nuanced mechanisms of community turnover (Figure 5). HP was the most significant contributor to functional composition changes across unprotected areas (Figure 5), indicating that widespread human activity outside PAs in mountain forests had a profound impact on the assemblage processes of medium- and large-sized mammals (Li, Bleisch, et al., 2024). Our results also identified HM as the most important driver of total functional β diversity and the components of these assemblages (Figures 4 & 5), highlighting the fragility of mountain ecosystems in the Anthropocene (Rahbek et al., 2019). Taken together, our findings are consistent with previous studies that show anthropogenic disturbances represent the main threat to global biodiversity (Matthews et al., 2024; Rodríguez-Caro et al., 2023).

We found that PAs in the Hengduan Mountains, southwest China, play a crucial role in safeguarding the functional diversity of medium- and large-sized mammal assemblages. Because differences in regional species pools and habitat conditions can confound the perceived effectiveness of PAs (Brodie et al., 2023), rigorous assessments must compare temporal trends inside and outside park boundaries. We therefore recommend coordinated, long-term monitoring in protected and unprotected sites across the range to refine understanding of mammalian functional diversity and the ecological processes it supports. Variation in functional diversity among sites was driven primarily by differences in species richness, suggesting that conservation actions may yield the greatest returns if they prioritize species-rich locations while still addressing the needs of lower-diversity areas (Socolar et al., 2016). Human pressure emerged as the strongest predictor of functional β diversity, underscoring the vulnerability of mountain ecosystems in the Anthropocene (Rahbek et al., 2019). Quantifying and mitigating these pressures inside and around PAs are therefore urgent if conservation targets are to be met and ecosystem functions and services are to be sustained over the long term (Jones et al., 2018).

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DATA AVAILABILITY STATEMENT

All data together with the R code used for analyses are openly available in Science Data Bank: <https://doi.org/10.57760/sciencedb.25286>.

ORCID

Xueyou Li  <https://orcid.org/0000-0002-4705-6082>

Wenqiang Hu  <https://orcid.org/0009-0006-2576-3890>

Kenneth Otieno Onditi  <https://orcid.org/0000-0003-4034-6818>

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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